The Integumentary Engine

Hull Breach Protocols: Burn Hydraulics, Structural Corrosion, & Bio-Hazard Containment

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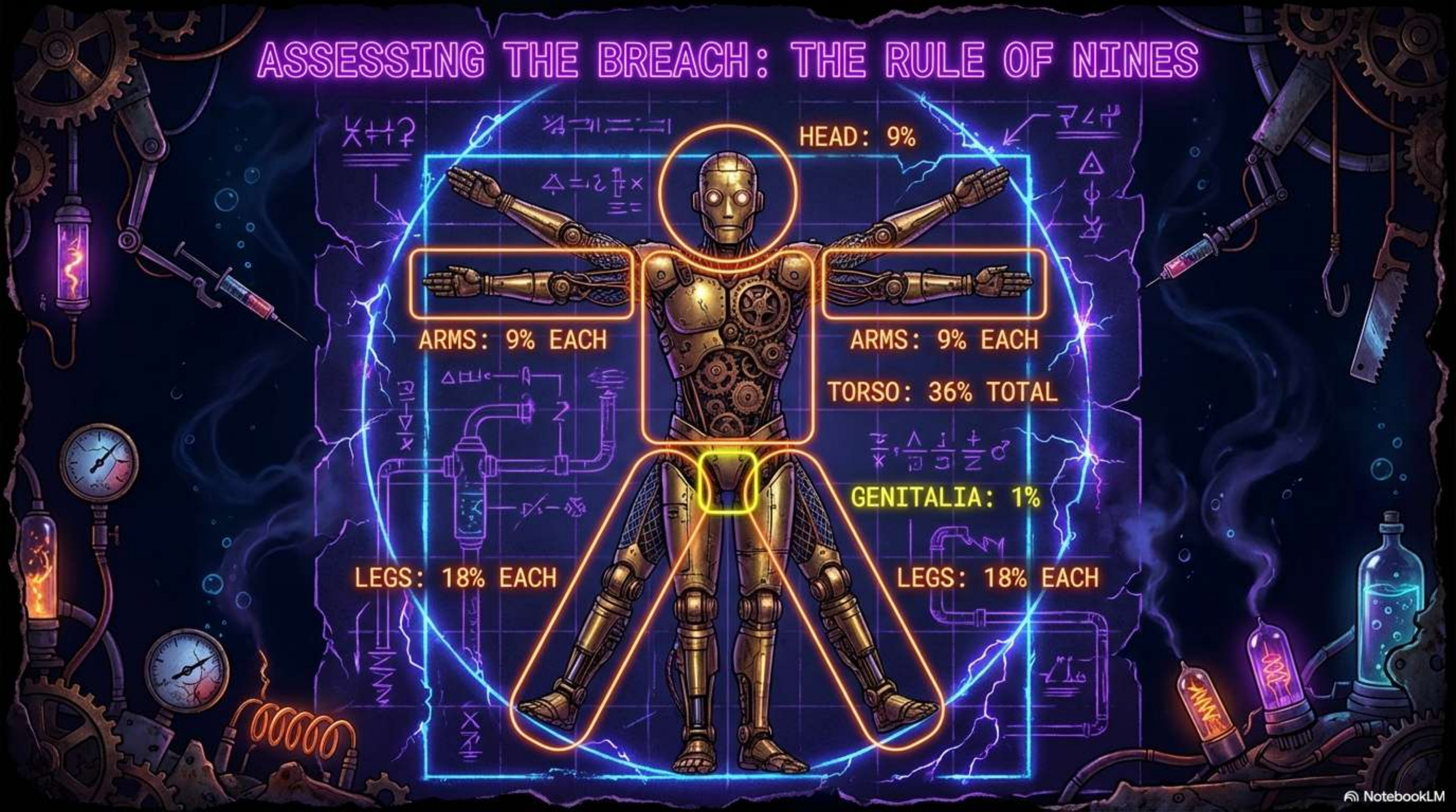
ASSESSING THE BREACH: THE RULE OF NINES
HEAD: 9%
ARMS: 9% EACH
ARMS: 9% EACH
TORSO: 36% TOTAL
GENITALIA: 1%
LEGS: 18% EACH
LEGS: 18% EACH

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SYSTEM ALERT: HYDRAULIC FAILURE (SHOCK PHASE)

FLUID SHIFT:
Bloodstream >>
Interstitial Space

PRIORITY #1: Hypovoleemic Shock

PRIORITY #1: Hypovolemic Shock
Hyperkalemia (High K+)
Metabolic Acidosis

Signs:
Tachycardia, Hypotension,
Clammy Skin

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CRITICAL INTAKE FAILURE: RESPIRATORY DISTRESS

EMERGENCY ACTIONS

1. RAISE HEAD OF BED
2. ADMINISTER OXYGEN (10L Mask)
3. PREPARE FOR INTUBATION

O2 SATURATION < 93% = CRITICAL LEVEL D

Assessment: Soot, Singed Hair, Stridor, Laryngeal Edema

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Rye REFUELING PROTOCOL: THE PARKLAND FORMULA

$$4\ \text{mL} \times \left(\begin{matrix}\text{Weight}\\ \text{kg}\end{matrix}\right) \times \left(\begin{matrix}\text{\% TBSA}\\ \text{Burned}\end{matrix}\right) = \begin{matrix}\text{24hr FLUID}\\ \text{TOTAL}\end{matrix}$$

FIRST 8 HOURS: INFUSE 50% OF TOTAL

NEXT 16 HOURS: INFUSE REMAINING 50%

EXAMPLE: 70kg x 50% Burn = 14,000 mL Total.
(Give 7,000 mL in first 8 hours).

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SYSTEM OUTPUT & MEDICATION

TARGET URINE OUTPUT:
30 - 50 mL/hour

🚨 🚨 WARNING: 🚨 🚨
< 30 mL/hr = RENAL FAILURE

MEDICATION PROTOCOLS

1. TETANUS TOXOID
   (Passive Immunity > 5yrs)
2. SILVADENE
   (Topical Antimicrobial)
3. PAIN MGMT
   (IV Only - No IM/Oral)

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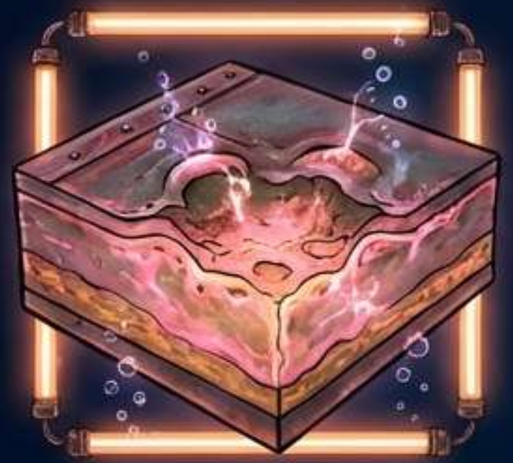
THE CORROSION MATRIX: ULCER STAGING

STAGE 1

STAGE 2



STAGE 3

STAGE 4

STAGE 1

Non-blanchable redness. Hull intact.

STAGE 2

Partial thickness. Blister/Crater. Pink dermis.

STAGE 3

Full thickness. Yellow subcutaneous fat visible.

STAGE 4

Full destruction. Muscle/Bone exposed.

UNSTAGEABLE if covered in Black Eschar.

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RISK CALIBRATION: THE BRADEN SCALE

Sensory Perception | Moisture | Activity | Mobility | Nutrition | Friction

SCORE 19-23 = LOW RISK

SCORE < 18 = HIGH RISK

Lower Number = Higher Maintenance Required.

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Rye

PREVENTATIVE MAINTENANCE PROTOCOLS

ROTATION:

- Turn q 2 Hours.
- Use Draw Sheet.
- Heel Protectors.

FUEL:

- High Calorie + High Protein Diet.

WARNING:

- NEVER massage reddened bony prominences.

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Rye

VIRAL BREACH: HERPES ZOSTER (SHINGLES)

Pathology: Reactivation of Varicella (Chicken Pox).

Action: STRICT ISOLATION / Contact Precautions.

Action: Negative Airflow if disseminated.

CAUTION: Nurses with no immunity/pregnancy must NOT enter.

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Psychotic Steampunk Triage

MICRO-INFESTATION PROTOCOLS

SCABIES (MITES)

Highly Contagious.
Contact Precautions.

Tx: Permethrin (Scabicide).
Wash linens in HOT water.

PEDICULOSIS (LICE)

Tx: Kwell Shampoo.
Remove Nits with fine comb.

Treat all family members.

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ENVIRONMENTAL THREATS

LYME DISEASE (Vector: Deer Tick)

Sign: Bullseye Rash

Tx: Doxycycline / Antibiotics

CONSTRICTION EMERGENCY

Action: Remove rings/bands IMMEDIATELY from injured extremities.

Risk: Ischemia / Necrosis.

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SURVEILLANCE: MALIGNANCY DETECTION

PREVENTION: SPF 30+ Sunscreen. Avoid Midday Sun.

THE ABCD SCAN

- **A** Asymmetry
- **B** Border Irregularity
- **C** Color Variation
- **D** Diameter > 6mm

Target: Melanoma / Basal Cell Carcinoma

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THE MASTER MECHANIC'S TRIAGE

KILLERS (DO NOW)

Roboto Mono

O2 Sat < 93% (Airway)

Stridor / Laryngeal Edema

Hypovolemic Shock

WARNINGS (MONITOR)

Roboto Mono

Stage 1-2 Ulcers

Fever (Non-fresh Post-op)

STABLE (LATER)

Roboto Mono

Chronic Skin Issues

Teaching / Discharge

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Rye

"MAINTAIN HULL INTEGRITY"

Rcuity First. Safety Always.

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